

Verification of Maximum Power Transfer Theorem	
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Roll Number: 23EE019	Date: 17-10-2024

Aim:

Verification of Maximum Power Transfer Theorem

Theory:

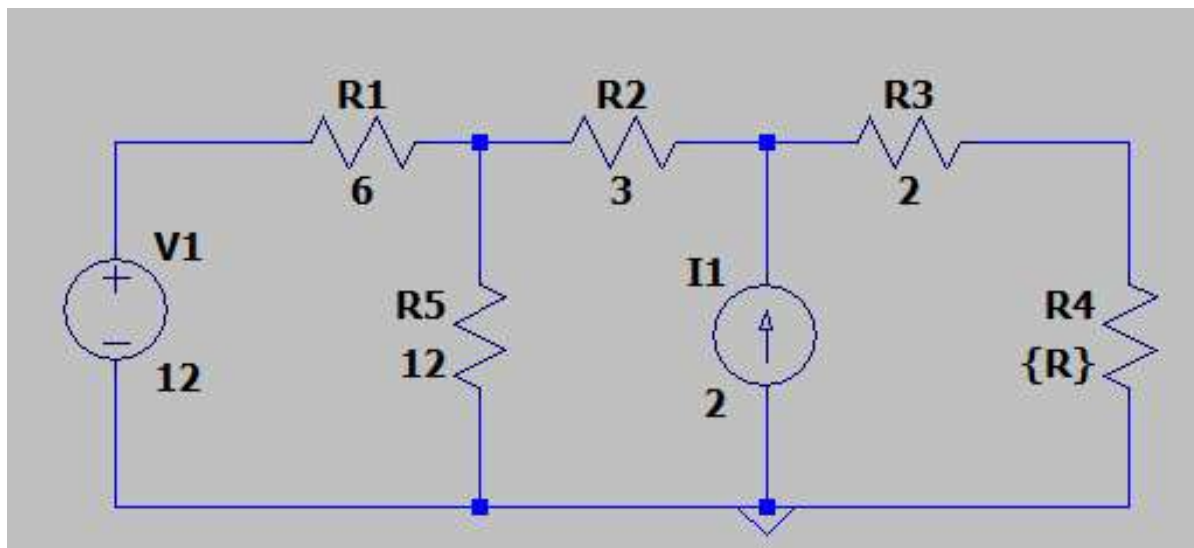
Maximum Power Transfer Theorem:

The maximum power transfer theorem states that the maximum power is delivered to a load when the load resistance is equal to the source resistance. This theorem can be applied to linear circuits that have one or more current or voltage sources and linear impedances.

Question 1.

Find the value of R_L for maximum power transfer in the circuit. Find the maximum power.

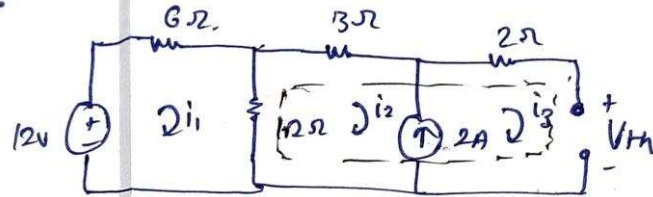
Circuit Diagram:



Theoretical Calculation:

Determination of Thevenin's Voltage V_{th} :

For V_{th} :-



$$i_3 + 2 = i_2 \Rightarrow i_3 = 0 \text{ A}$$

$$i_3 = 2 \text{ A} \quad i_2 = -2 \text{ A}$$

$$-12 + 6i_1 + 12i_1 - 12i_2 = 0$$

$$18i_1 - 12i_2 = 12 \quad (1)$$

$$18i_1 + 24 = 12$$

$$i_1 = \frac{12 - 24}{18}$$

$$i_1 = \frac{-12}{18} = -\frac{2}{3}$$

$$3i_1 - 12i_2 - 12i_1 + 3i_2 + 2i_3$$

$$i_1 = -\frac{2}{3} \text{ A}$$

$$12(2 - i_1) + 3i_2 + 2i_3 + V_{th} = 0$$

$$15i_2 - 12i_1 + 2i_3 = -V_{th}$$

$$15 \times 2 - (2 \times -\frac{2}{3}) + 2 \times 0$$

$$= 30 + 8 = 22 \text{ V}$$

$$V = 22 \text{ V}$$

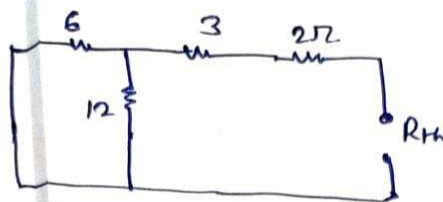
$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

$$= \frac{(22)^2}{4 \times 9} = 13.44$$

$$P_{max} = 13.44 \text{ W}$$

Determination of Thevenin's Resistance R_{th} :

For R_{th} :-



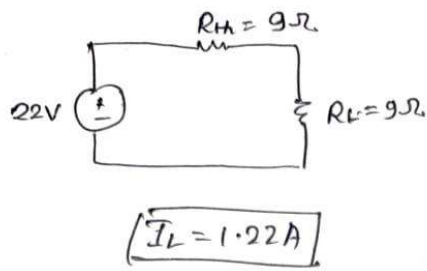
$$R_{th} = \frac{6 \times 12}{6 + 12} + 3 + 2$$

$$R_{th} = \frac{6 \times 12}{18} + 3 + 2$$

$$R_{th} = 9 \Omega$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Ques 1 Thevenins equivalent Circuit :-



If $R_L = R_{th} \Rightarrow$ Maximum power Transfer

$$I_L = \frac{V}{R_L + R_{th}} = \frac{22}{9 + 9} = 1.22A$$

Maximum Power Delivered to Load R_L :

$$P_{max} = \frac{V_{th}^2}{4 R_{th}}$$

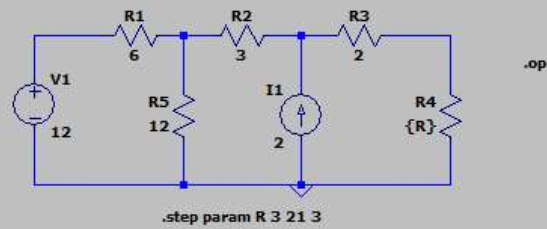
$$= \frac{(22)^2}{4 \times 9} = 13.44$$

$$P_{max} = 13.44 W$$

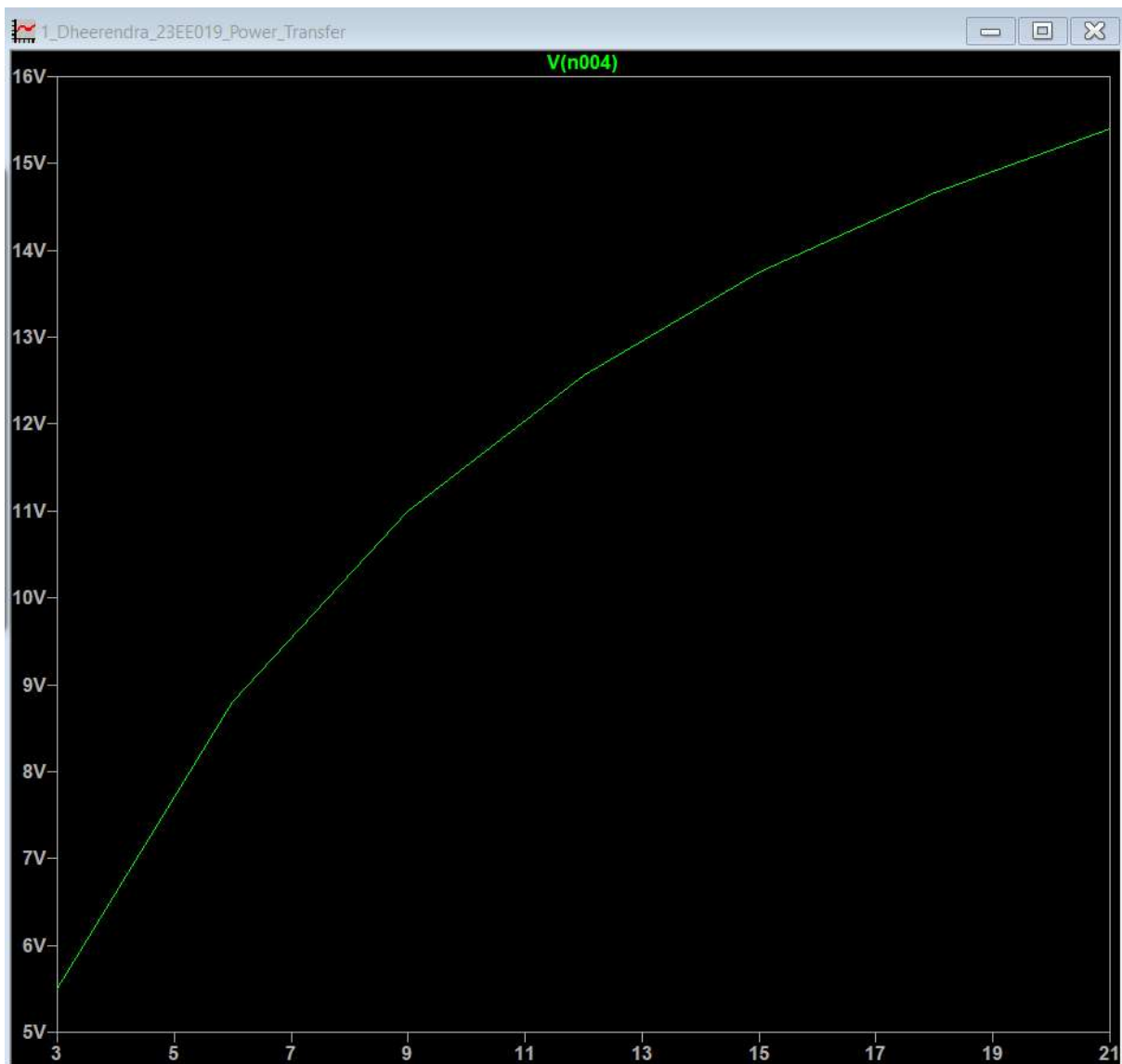
Verification by Simulation:

Circuit Diagram:

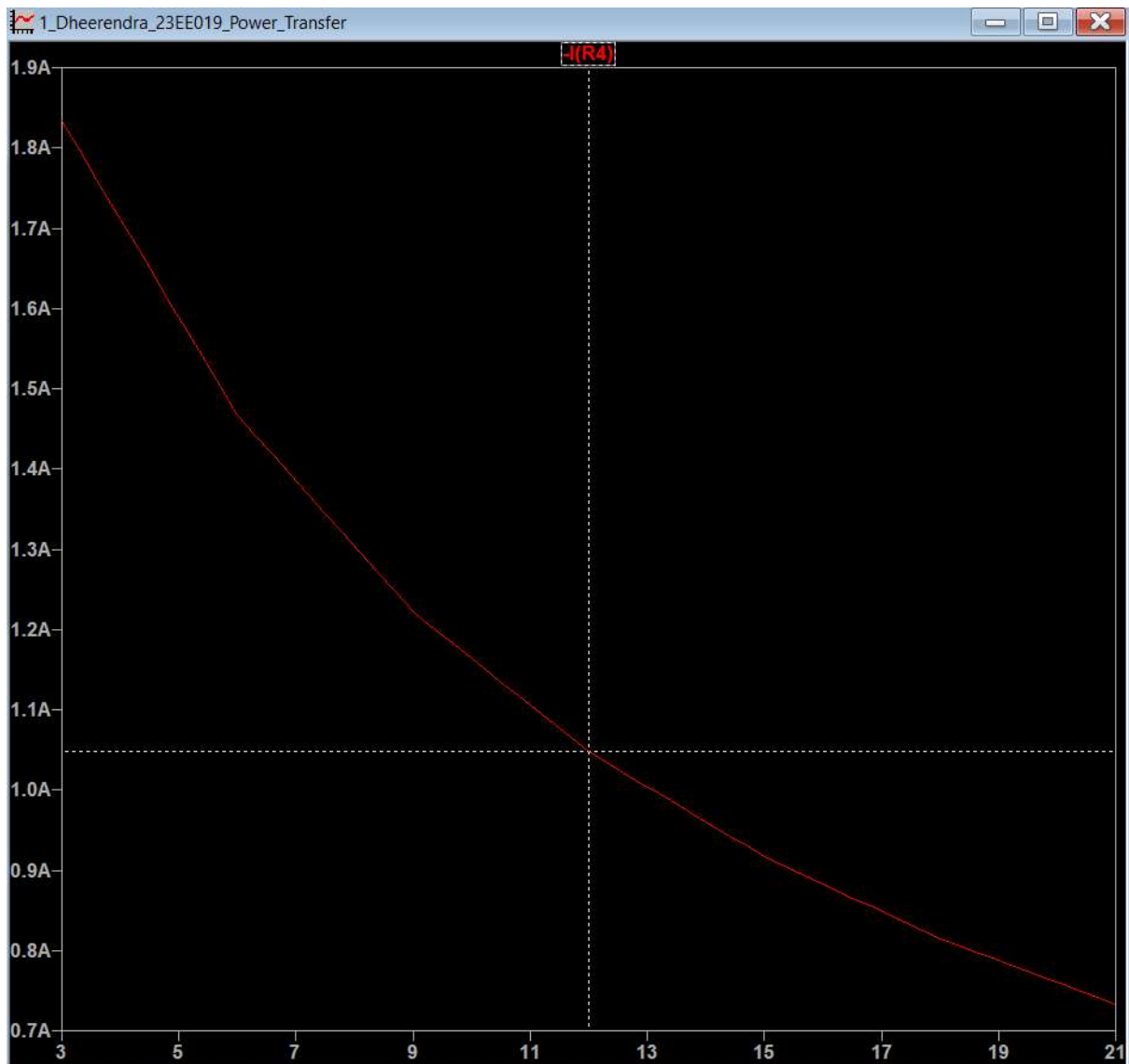
Que 1. Find the value of R_L for maximum power transfer in the circuit. Find the maximum power.



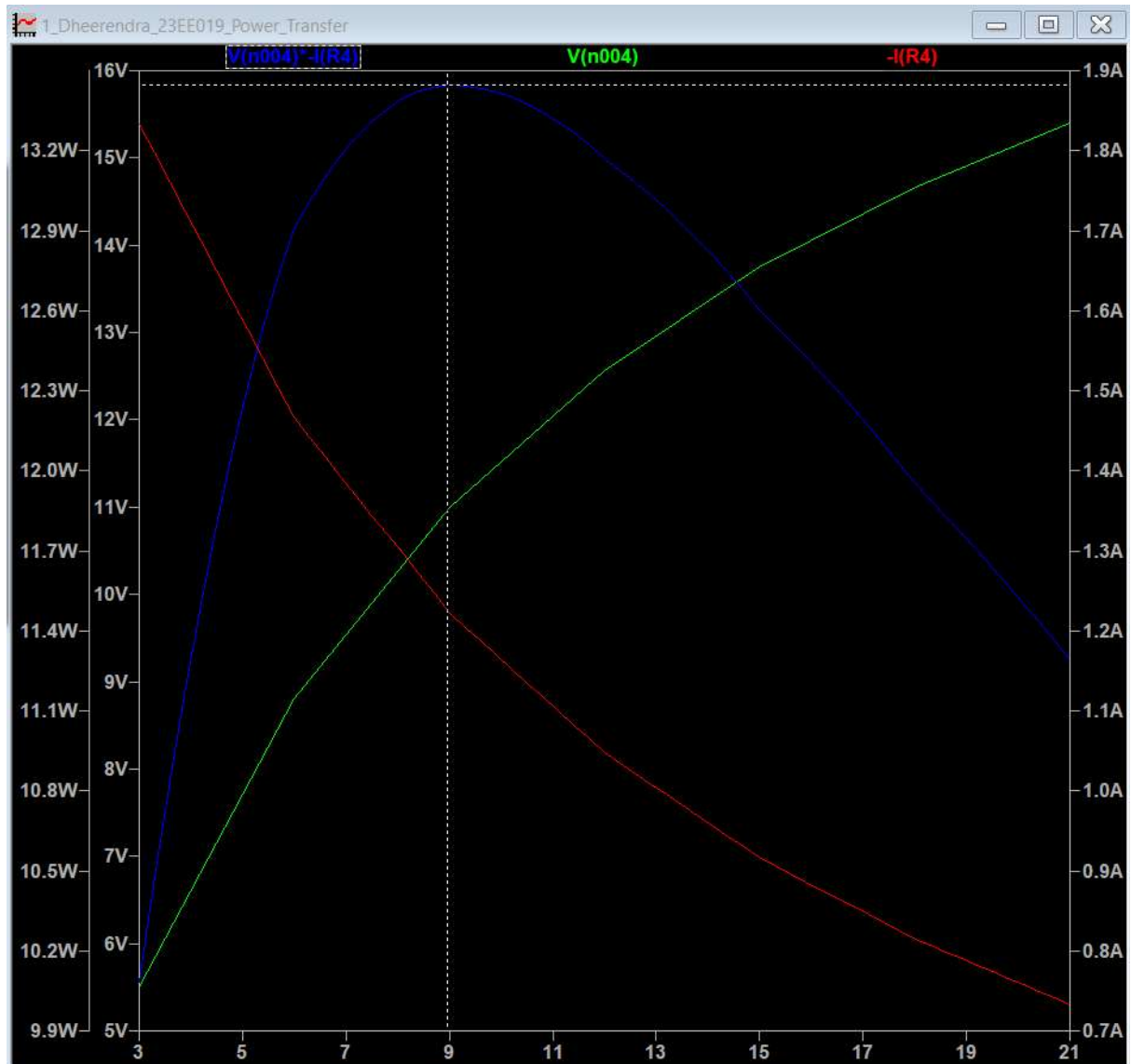
Plot of Voltage across R_L Voltage:



Plot of Current across R_L :



Determination of Maximum Power delivered to load R_L :



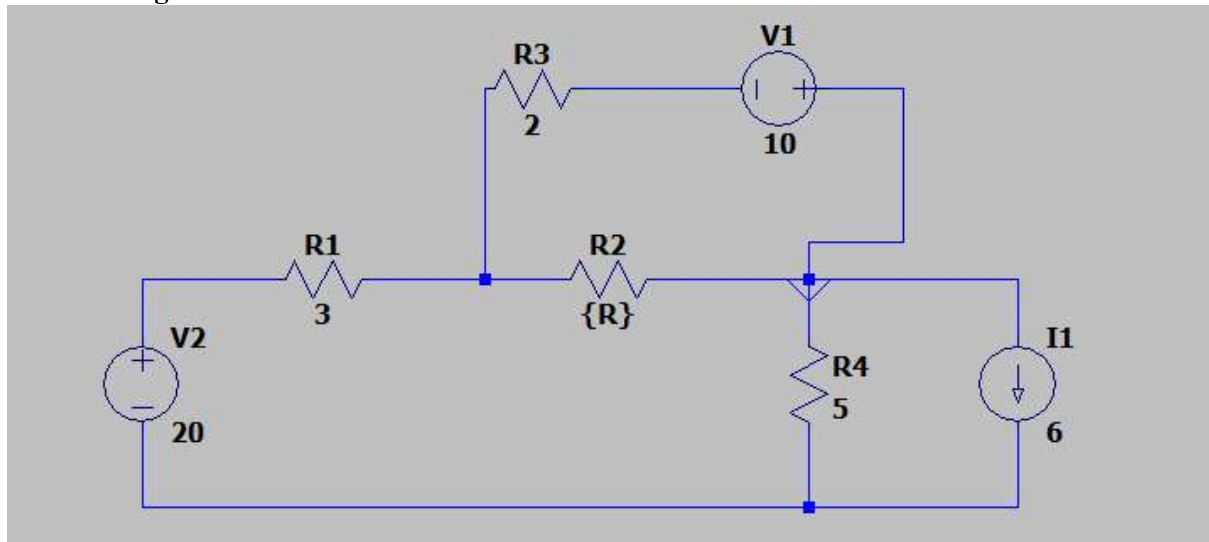
Observation Table:

S. No.	V_{th} (V)	$R_L = R_{th}$ (Ω)	I_L (A)	Power through Load Resistance $P_{L(max)} = I_L^2 \cdot R_L$ (W)	
1.	22	9	1.221776	Practical Value	Theoretical Value
				13.444443 W	13.44

Question 2.

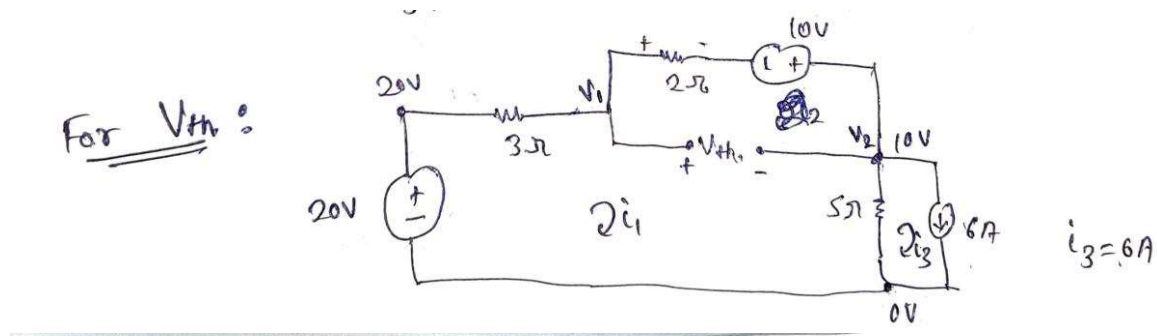
Find the maximum power that can be delivered to the resistor R in the circuit.

Circuit Diagram:



Theoretical Calculation:

Determination of Thevenin's Voltage V_{th} :



$$-20 + 3i_1 + 2i_1 - 10 + 5i_1 - 5i_3 = 0$$

$$-30 + 10i_1 - 30 = 0$$

$$i_1 = \frac{60}{10} = 6A$$

$$V_{th} = 2V$$

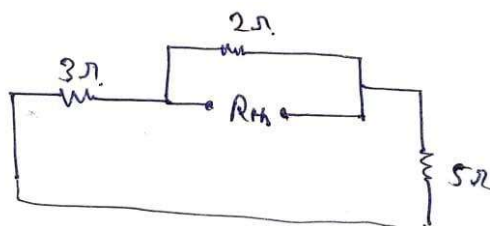
$$V_{2\Omega} = 2 \times 6 = 12V$$

$$V_{th} = V_{2\Omega} - 10$$

$$V_{th} = 12 - 10 = 2V$$

Determination of Thevenin's Resistance R_{th} :

For R_{th} :



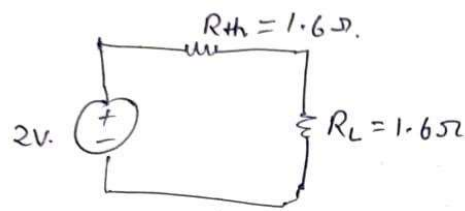
$$R_{th} = 3 +$$

$$R_{th} = \frac{8 \times 2}{8 + 2} = \frac{16}{10} = 1.6 \Omega$$

$$R_{th} = 1.6 \Omega$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Que 2 Thevenin's Equivalent circuit :-



If $R_L = R_{th} \Rightarrow$ Maximum power Transfer

$$I_L = \frac{V}{R_L + R_{th}} = \frac{2}{1.6 + 1.6} = 0.625A$$

$$\boxed{I_L = 0.625A}$$

Maximum Power Delivered to Load R_L :

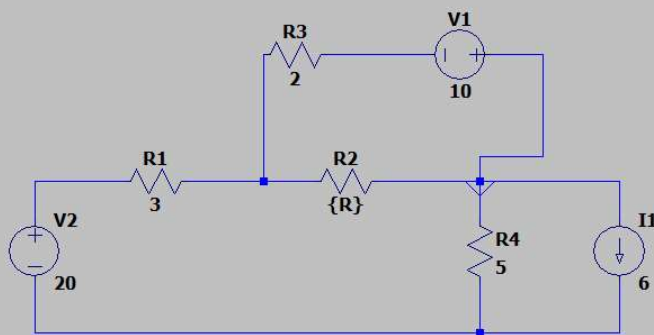
$$P_{max} = \frac{(V_{th})^2}{4R_{th}} = \frac{2^2}{4 \times 1.6} = \frac{4}{6.4} = 0.625W$$

$$\boxed{P_{max} = 0.625W}$$

Verification by Simulation:

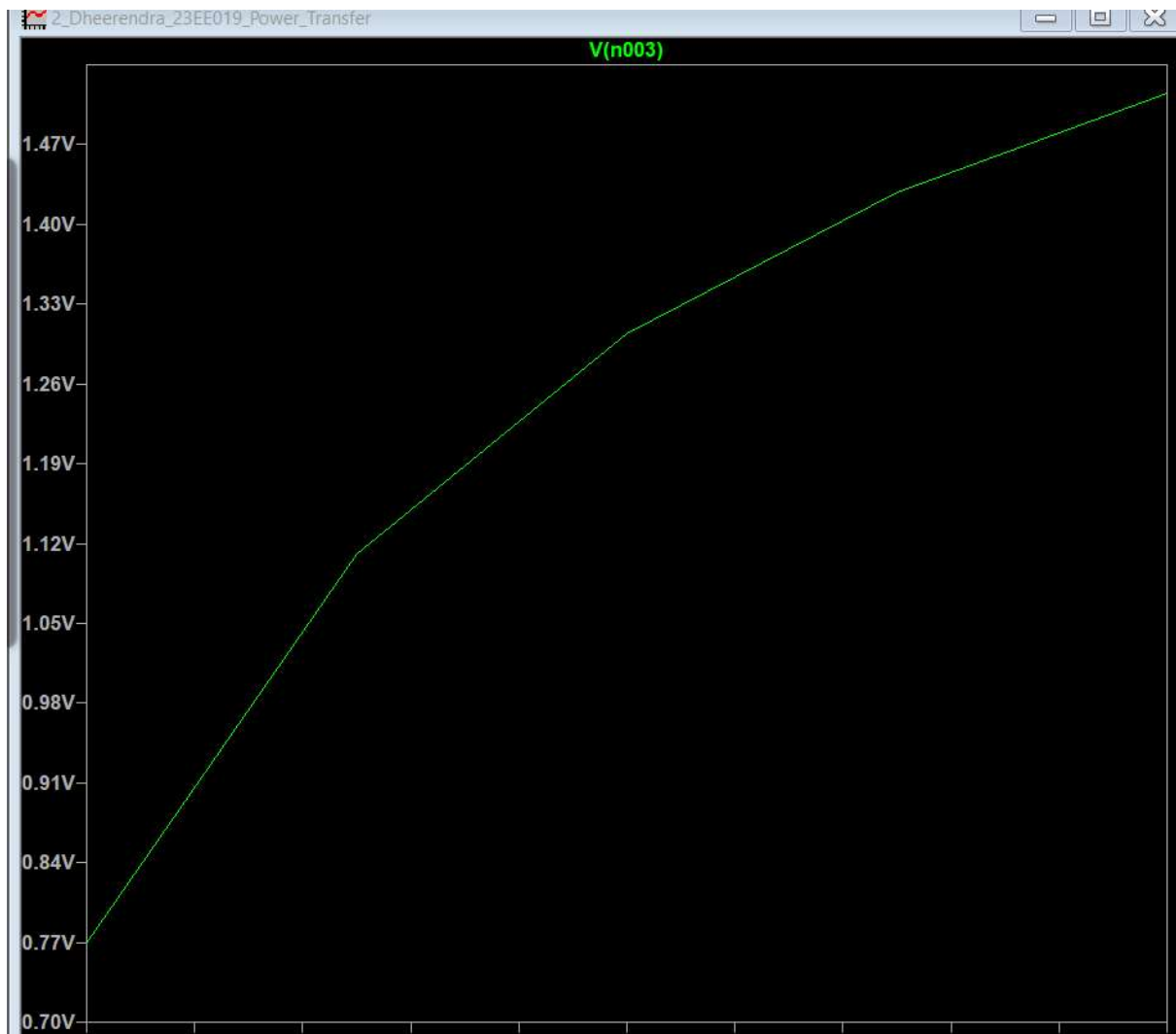
Circuit Diagram:

Que 2. Find the maximum power that can be delivered to the resistor R in the circuit.

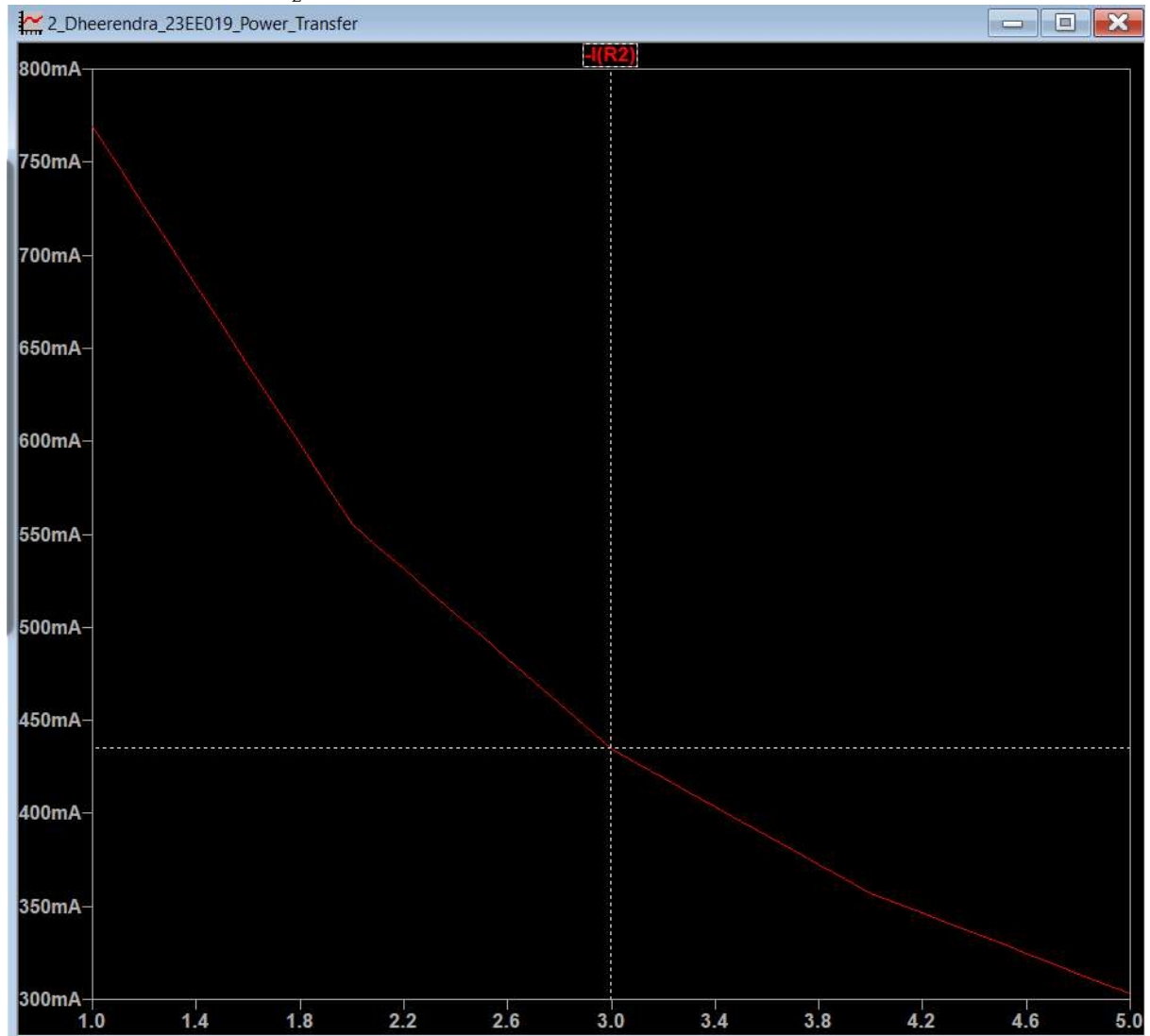


.op
.step param R 1 5 1

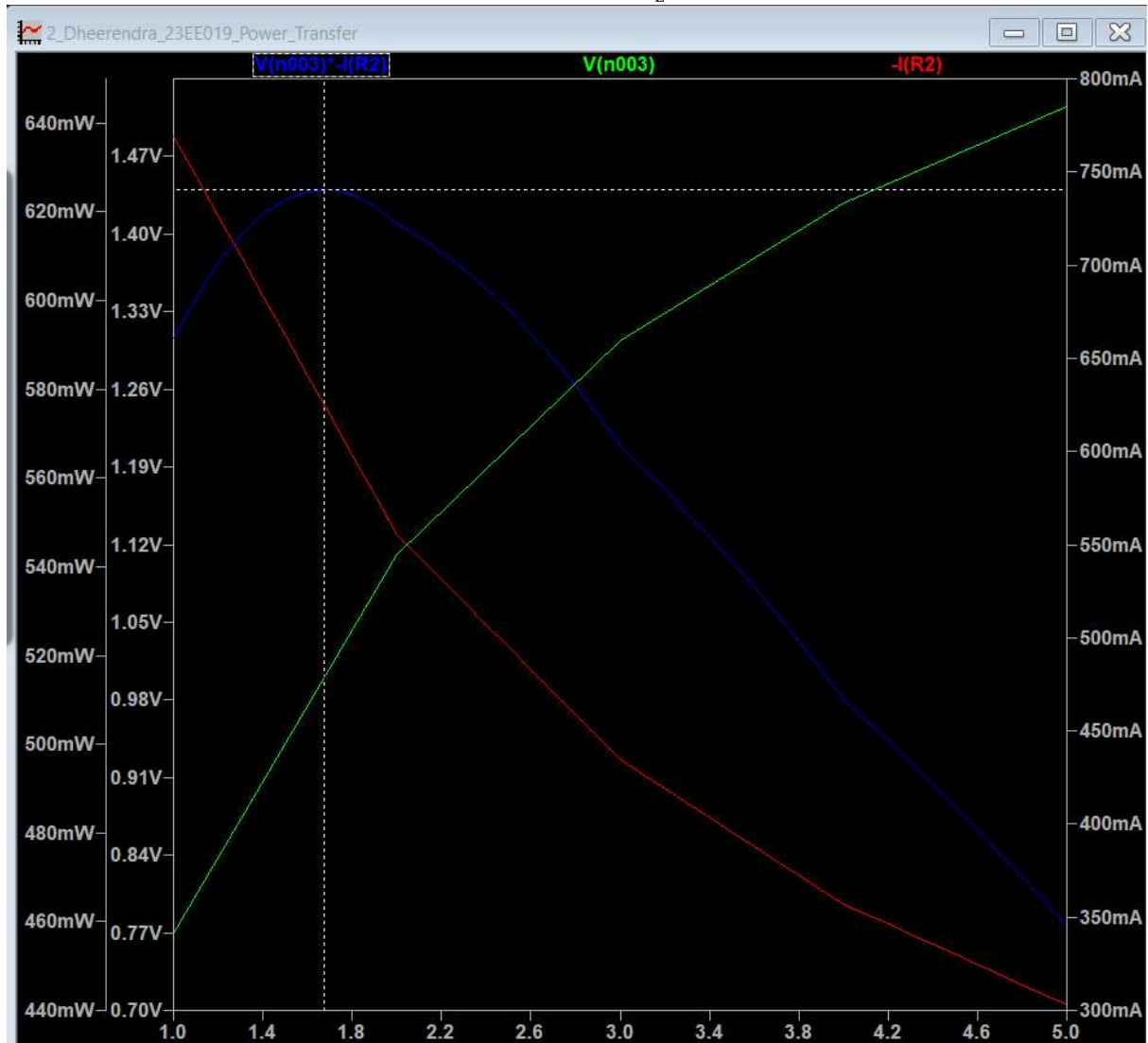
Plot of Voltage across R_L Voltage:



Plot of Current across R_L :



Determination of Maximum Power delivered to load R_L :



Observation Table:

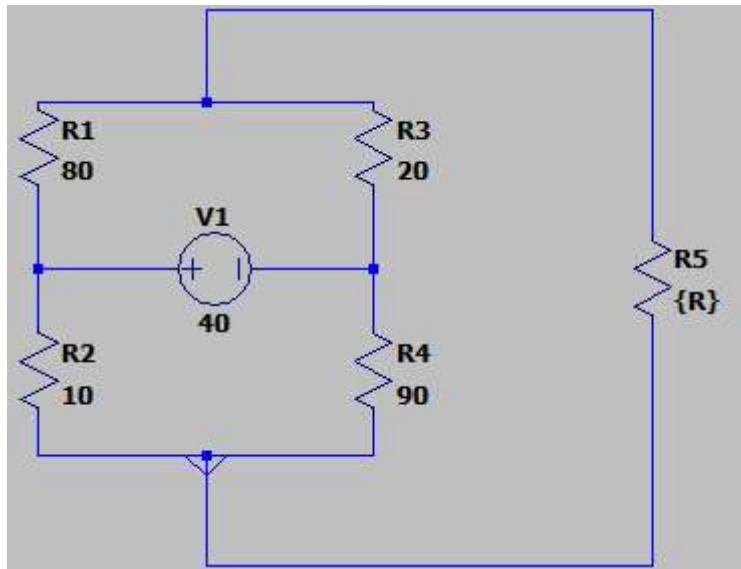
S. No.	V_{th} (V)	$R_L = R_{th}$ (Ω)	I_L (A)	Power through Load Resistance $P_{L(max)} = I_L^2 \cdot R_L$ (W)	
1.	2	1.6	0.625	Practical Value	Theoretical Value
				0.62500004	0.625

Question 3.

The variable resistor R in circuit is adjusted until it absorbs the maximum power from the circuit.

- Calculate the value of R for maximum power.
- Determine the maximum power absorbed by R .

Circuit Diagram:



Theoretical Calculation:

Determination of Thevenin's Voltage V_{th} :

For V_{th} :-

$$I = \frac{40}{50} = \frac{4}{5} \text{ A}$$

$$I' = \frac{4}{5} \times \frac{1}{2} = \frac{4}{10} = 0.4 \text{ A}$$

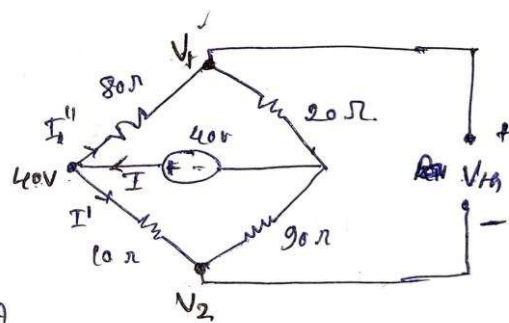
$$I'' = \frac{4}{10} = 0.4 \text{ A}$$

$$V_{80\Omega} = \frac{4}{10} \times 80 = 32 \text{ V}$$

$$V_{10\Omega} = \frac{4}{10} \times 10 = 4 \text{ V}$$

$$V_{th} = -V_2 + V_1 = -36 + 8$$

$$V_{th} = \underline{\underline{-28 \text{ V}}}$$



$$40 - V_1 = V_{80\Omega}$$

$$40 - V_1 = 32$$

$$V_1 = 40 - 32$$

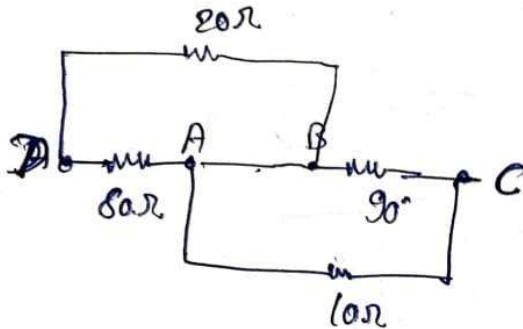
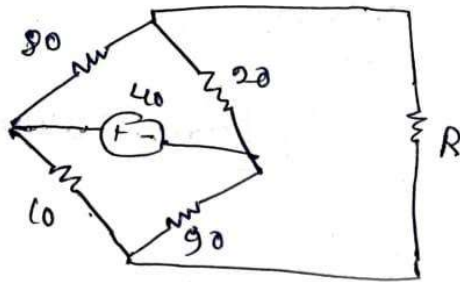
$$V_1 = 8 \text{ V}$$

$$40 - V_2 = V_{10\Omega}$$

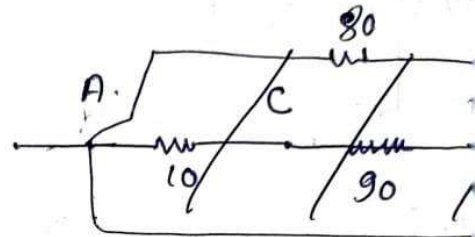
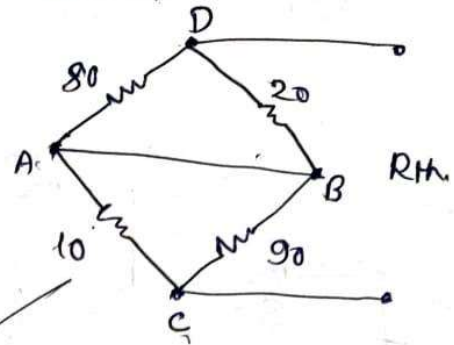
$$40 - 4 = V_2 \Rightarrow \boxed{V_2 = 36}$$

Determination of Thevenin's Resistance R_{th} :

Ques 3



For Rth:

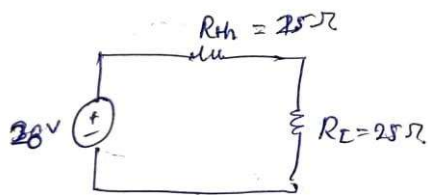


$$R_{th} = \frac{80 \times 20}{80 + 20} + \frac{90 \times 10}{90 + 10} = 16 + 9 = 25 \Omega$$

$$\boxed{R_{th} = 25 \Omega}$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Ques 3 Thevenin's Equivalent Circuit :-



if $R_L = R_{th} \Rightarrow$ Maximum power transfer

$$I_L = \frac{V}{R_L + R_{th}} = \frac{28}{25 + 25} = 0.56 \text{ A}$$

$$\boxed{I_L = 0.56 \text{ A}}$$

Maximum Power Delivered to Load R_L :

$$P_{max} = \frac{(V_{th})^2}{4R_{th}} = \frac{(-28)^2}{4 \times 25} = 7.84 \text{ W.}$$

$$\boxed{P_{max} = 7.84 \text{ W}}$$

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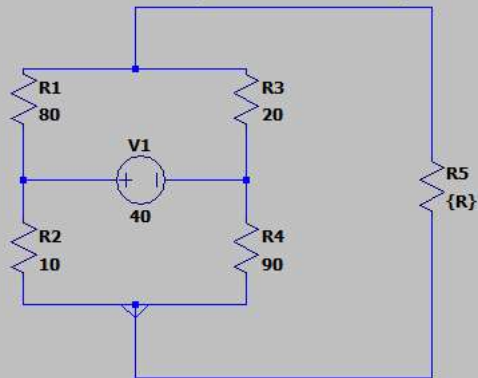
Verification by Simulation:

Circuit Diagram:

Que3. The variable resistor R in circuit is adjusted until it absorbs the maximum power from the circuit.

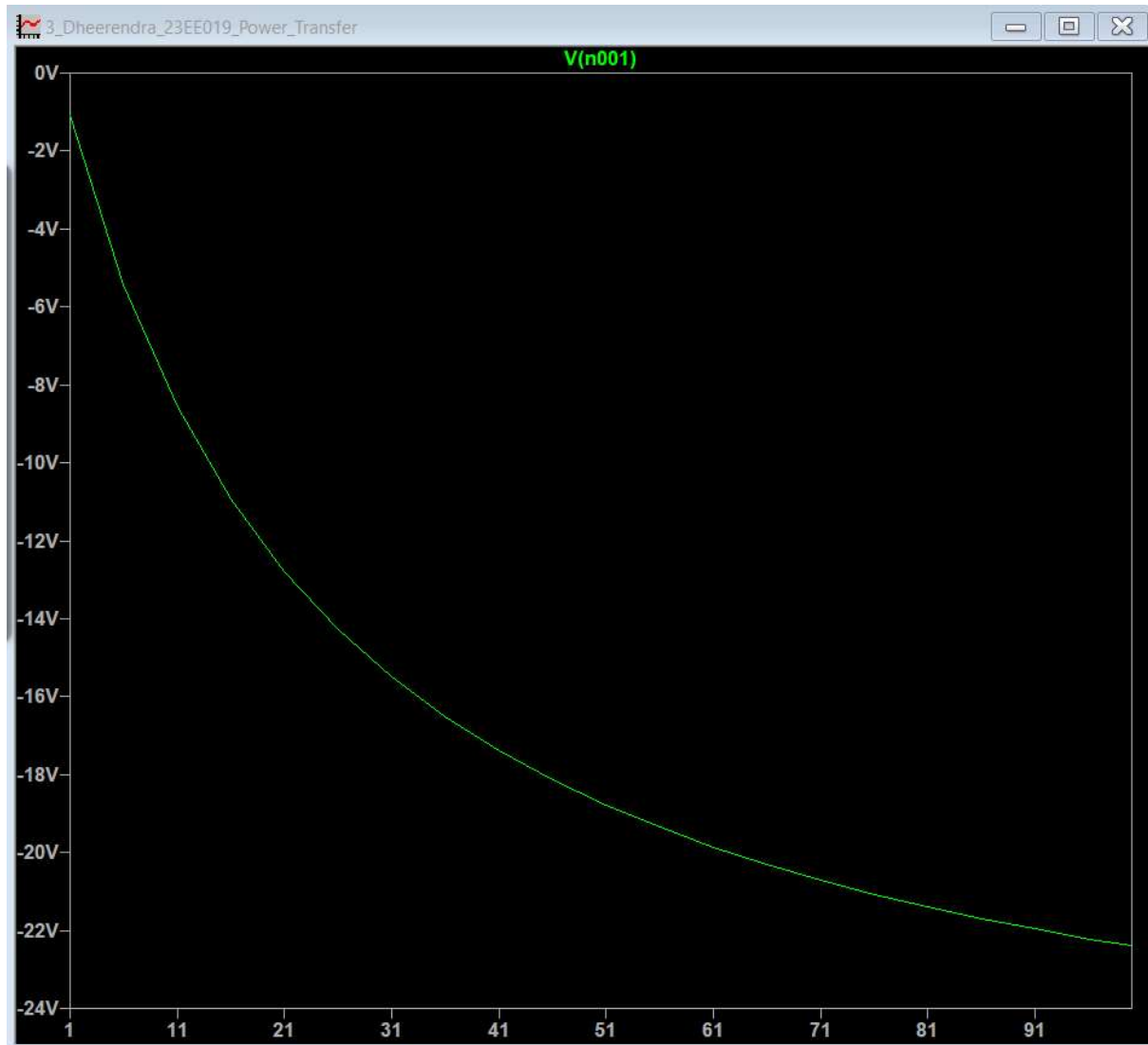
(a) Calculate the value of R for maximum power.

(b) Determine the maximum power absorbed by R.

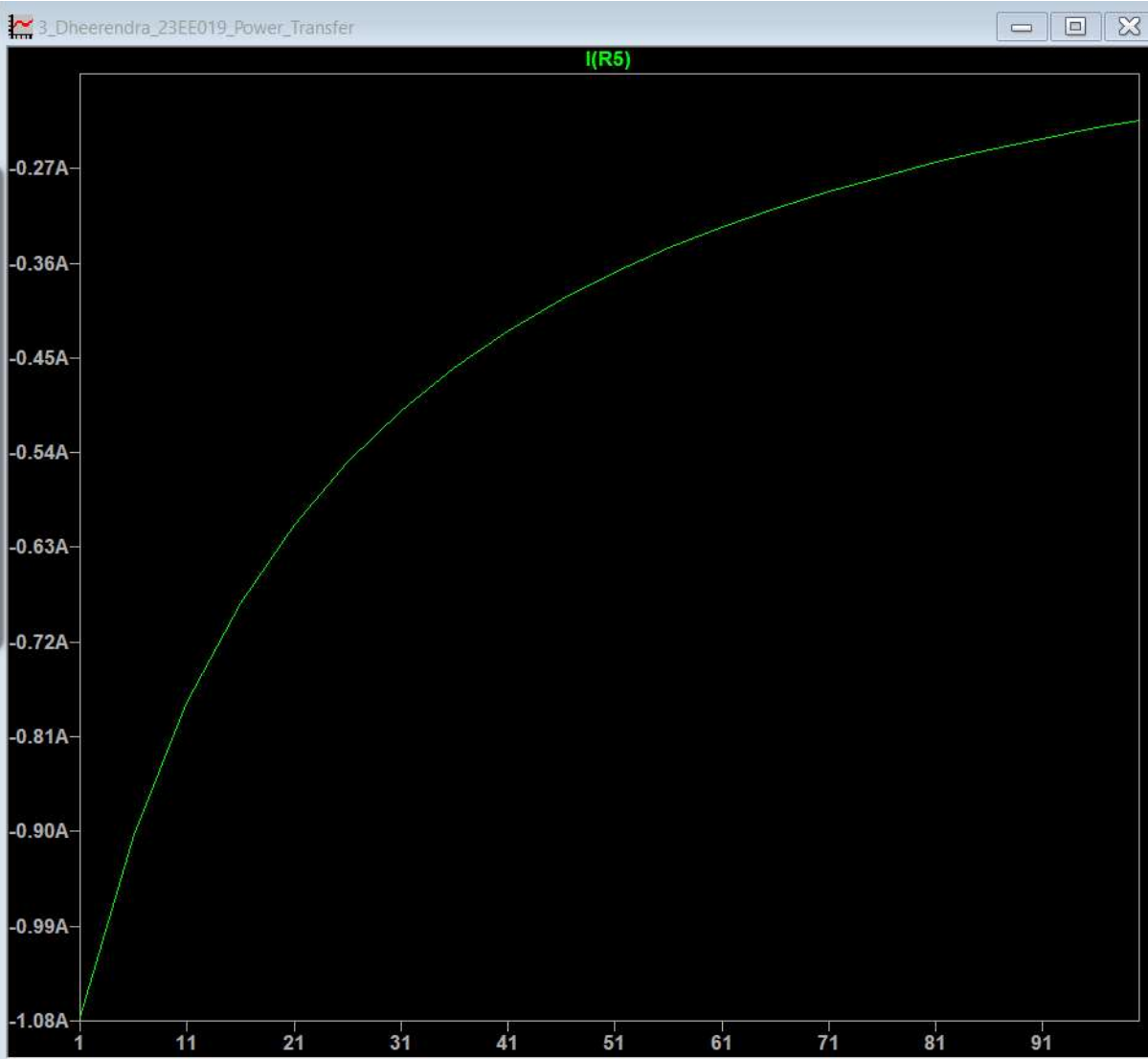


.op

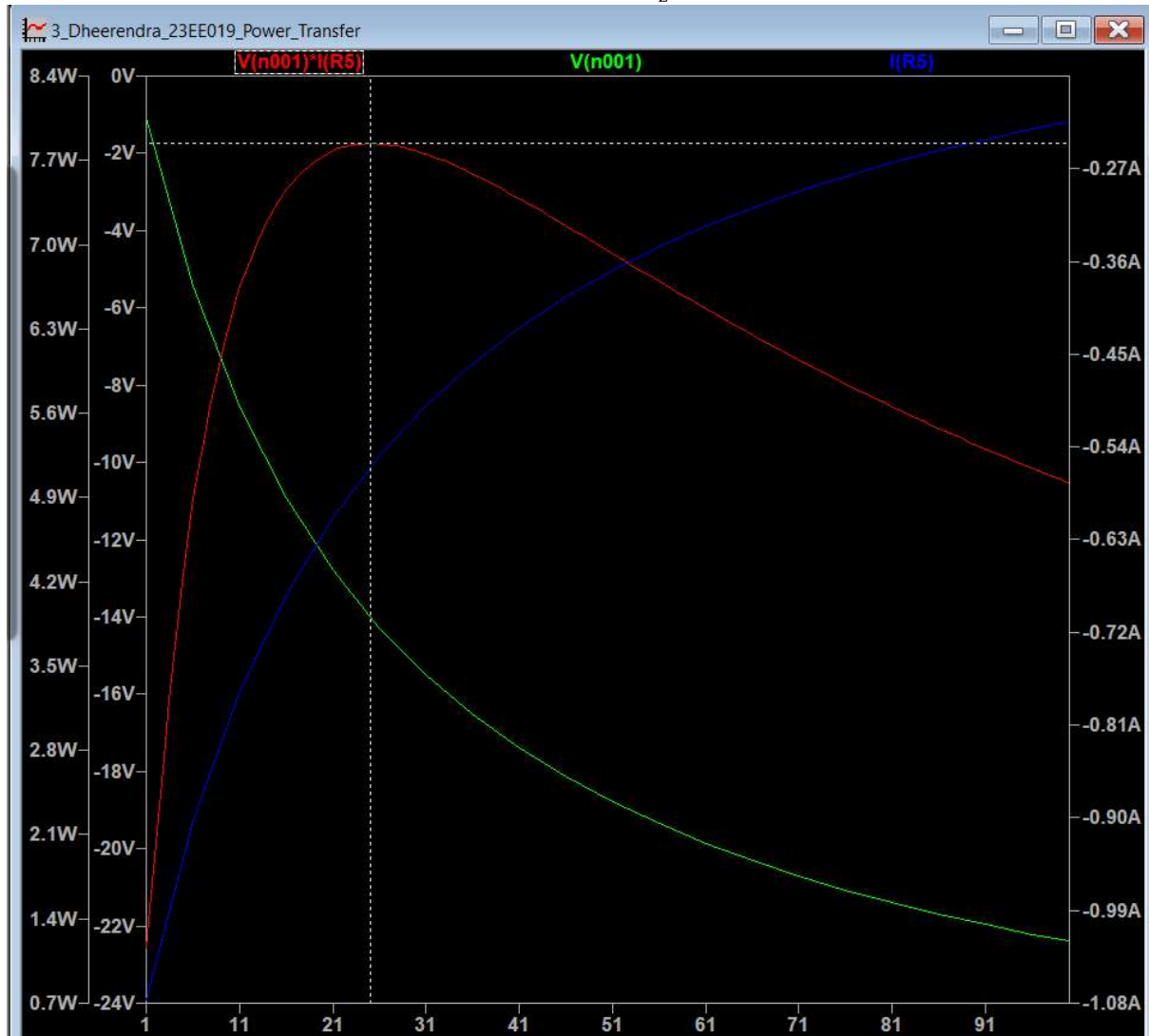
Plot of Voltage across R_L Voltage:



Plot of Current across R_L :



Determination of Maximum Power delivered to load R_L :



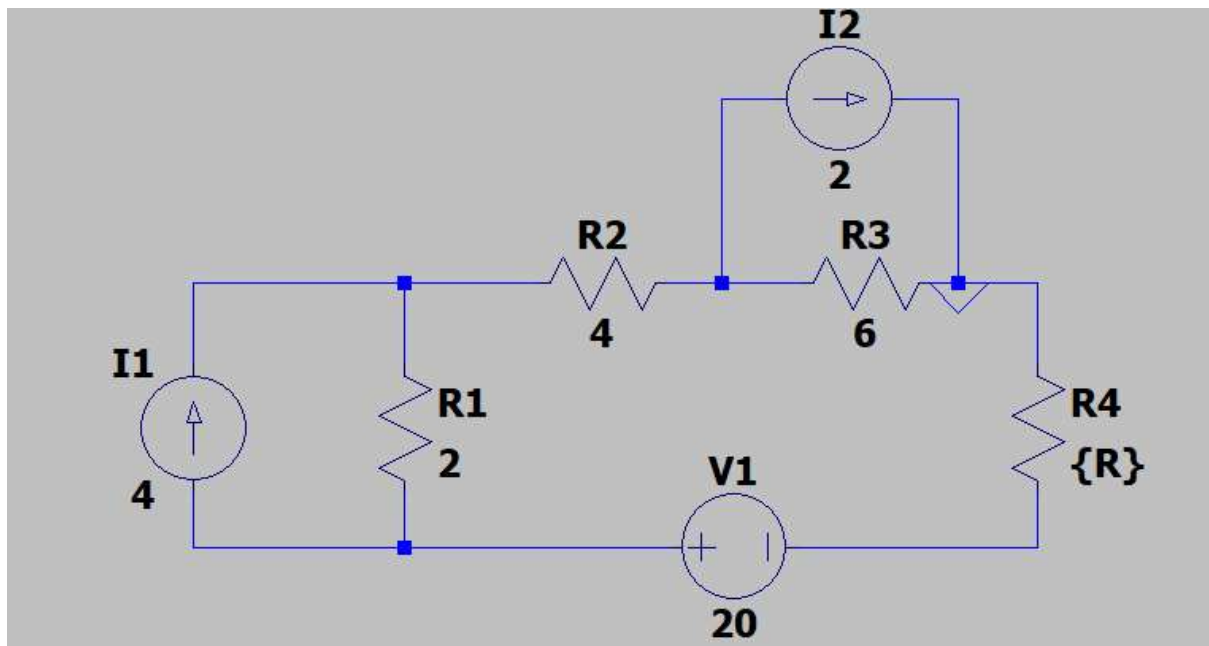
Observation Table:

S. No.	V_{th} (V)	$R_L = R_{th}$ (Ω)	I_L (A)	Power through Load Resistance $P_{L(max)} = I_L^2 * R_L$ (W)	
3.	28	25	0.56	Practical Value	Theoretical Value
				7.8399952	7.84

Question 4.

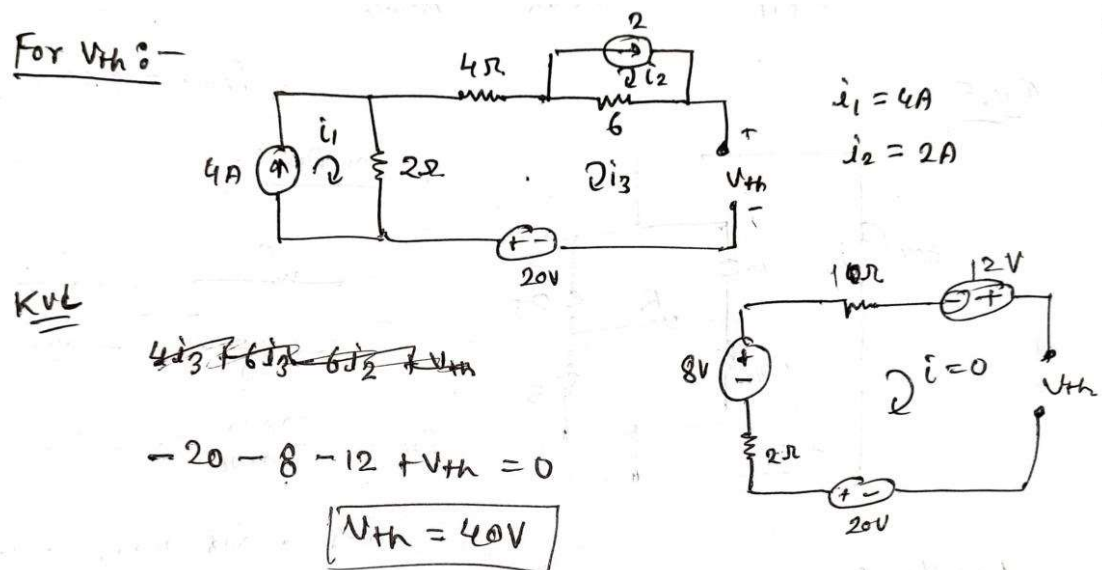
Find R_L for maximum power deliverable to R_L . Determine that maximum power.

Circuit Diagram:



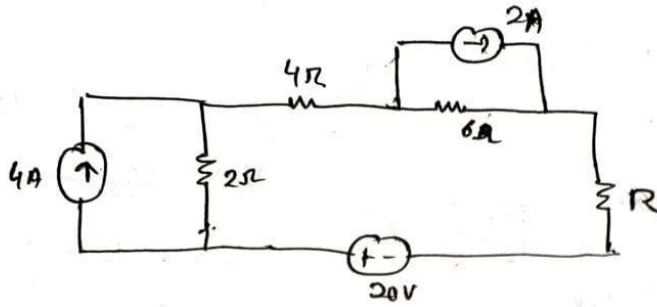
Theoretical Calculation:

Determination of Thevenin's Voltage V_{th} :

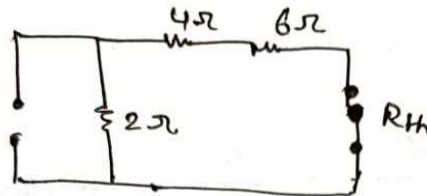


Determination of Thevenin's Resistance R_{th} :

Ques 4



For R_{th} :-

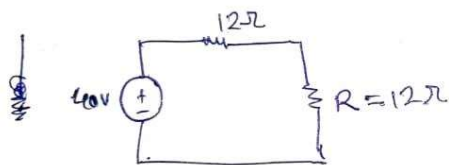


$$R_{th} = 6 + 4 + 2$$

$$R_{th} = 12\Omega$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Thevenin's equivalent circuit-



If $R_L = R_{th} \Rightarrow$ Maximum power transfer

$$I_L = \frac{V}{R_L + R_{th}} = \frac{40}{12 + 12} = \frac{40}{24}$$

$$I_L = 1.6667 A$$

Maximum Power Delivered to Load R_L :

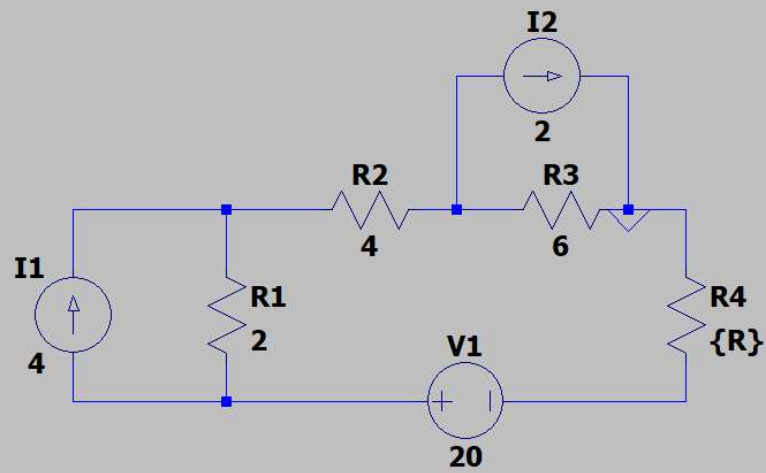
$$P_{max} = \frac{(V_{th})^2}{4R_{th}} = \frac{(40)^2}{4 \times 12} = 33.33 W$$

$$P_{max} = 33.33 W$$

Verification by Simulation:

Circuit Diagram:

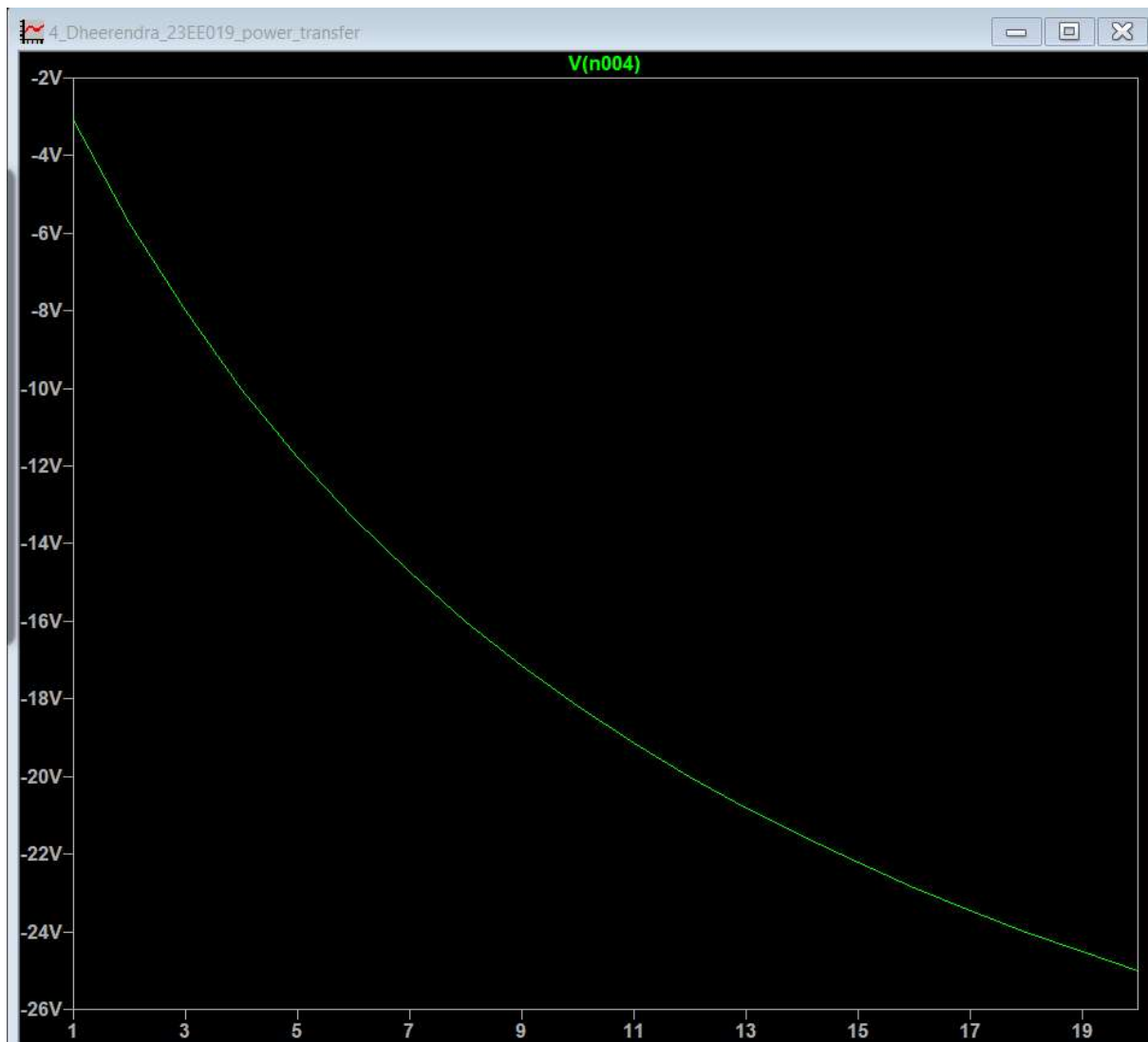
Que 4 Find R_L for maximum power deliverable to R_L . Determine that maximum power.



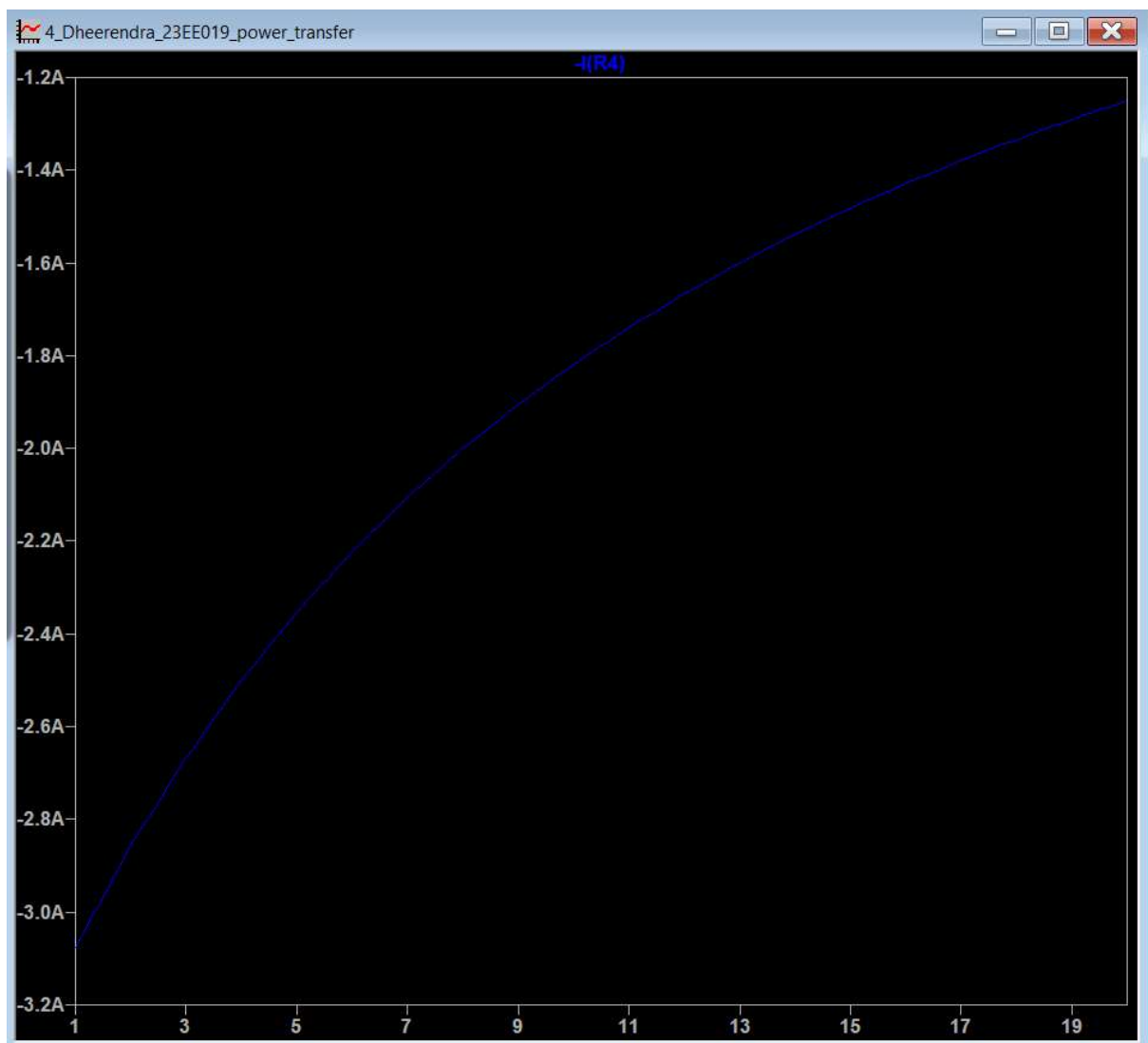
.step param R 1 20 1

.op

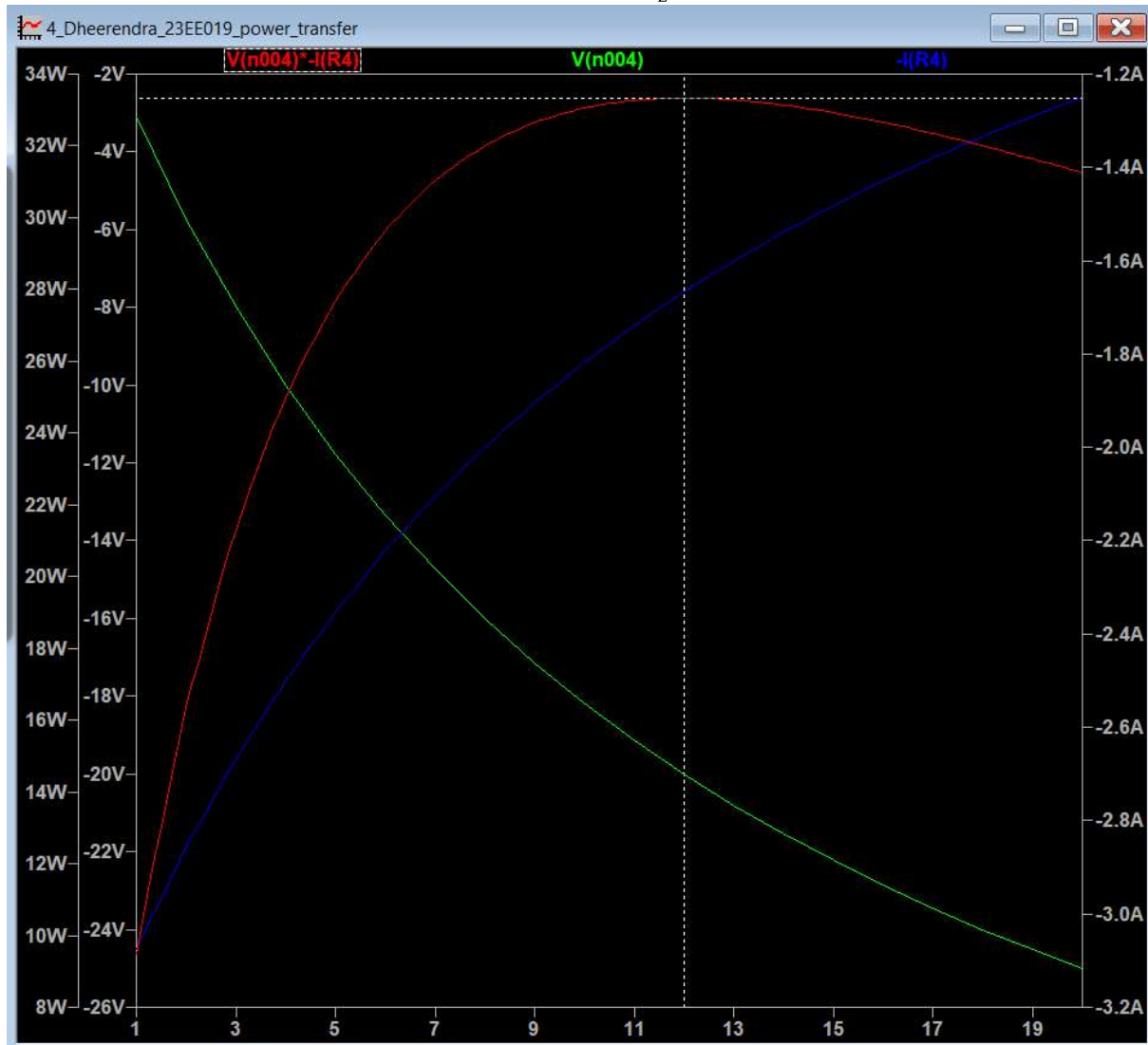
Plot of Voltage across R_L Voltage:



Plot of Current across R_L :



Determination of Maximum Power delivered to load R_L :



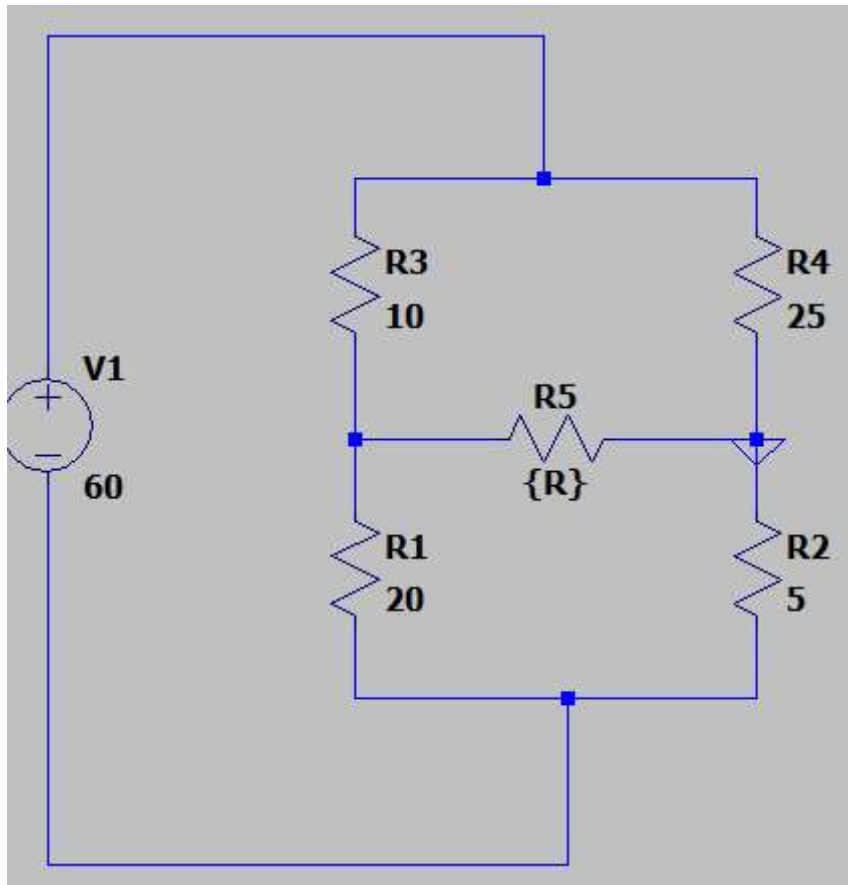
Observation Table:

S. No.	V_{th} (V)	$R_L = R_{th}$ (Ω)	I_L (A)	Power through Load Resistance $P_{L(max)} = I_L^2 \cdot R_L$ (W)	
4.	40	12	1.66667	Practical Value	Theoretical Value
				33.333332	33.33

Question 5.

Determine the maximum Power that can be delivered to the variable resistor R in the circuit

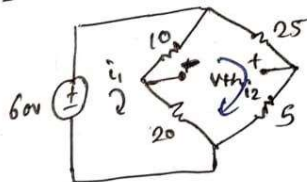
Circuit Diagram:



Theoretical Calculation:

Determination of Thevenin's Voltage V_{th} :

For V_{th} :



$$i_1 = \frac{60}{30} = 2A$$

$$i_2 = \frac{60}{30} = 2A$$

KVL $25i_2 + V_{th} - 10i_1 = 0$

$$\begin{aligned} V_{th} &= 10i_1 - 25i_2 \\ &= 20 - 50 = -30V \end{aligned}$$

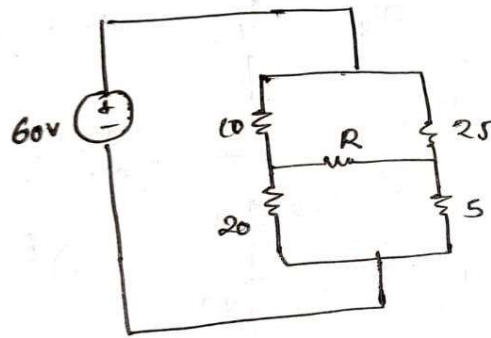
$$\boxed{V_{th} = -30V}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(-30)^2}{4 \times 10.933}$$

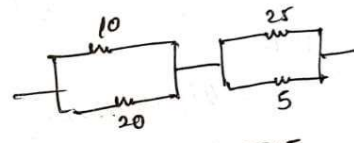
$$\boxed{P_{max} = 20.795W}$$

Determination of Thevenin's Resistance R_{th} :

Que. 5



For R_{th} :-



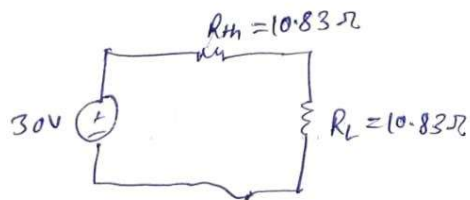
$$R_{th} = \frac{10 \times 20}{10 + 20} + \frac{25 \times 5}{25 + 5}$$

$$R_{th} = 6.667 + 4.167$$

$$\boxed{R_{th} = 10.83 \Omega}$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Que 5 Thevenin's Equivalent Circuit :-



If $R_L = R_{th} \Rightarrow$ Maximum power transfer.

$$I_L = \frac{V}{R_L + R_{th}} = \frac{30}{10.83 + 10.83} = 1.385 A$$

$$\boxed{I_L = 1.385 A}$$

Maximum Power Delivered to Load R_L :

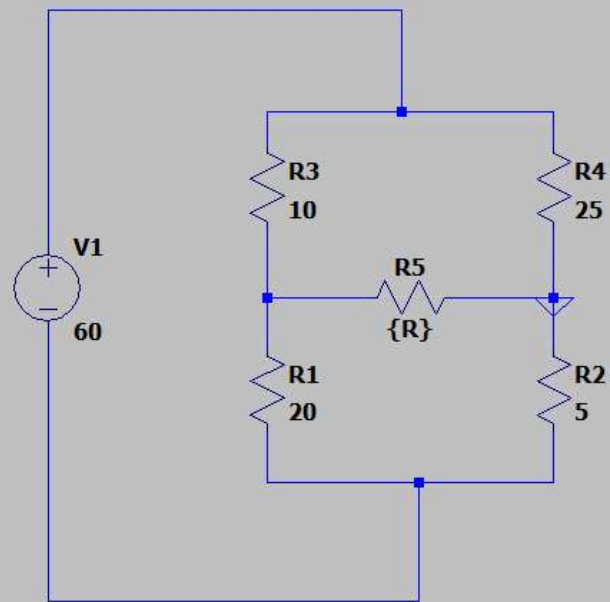
$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(30)^2}{4 \times 10833}$$

$$P_{max} = 20.795 \text{ W}$$

Verification by Simulation:

Circuit Diagram:

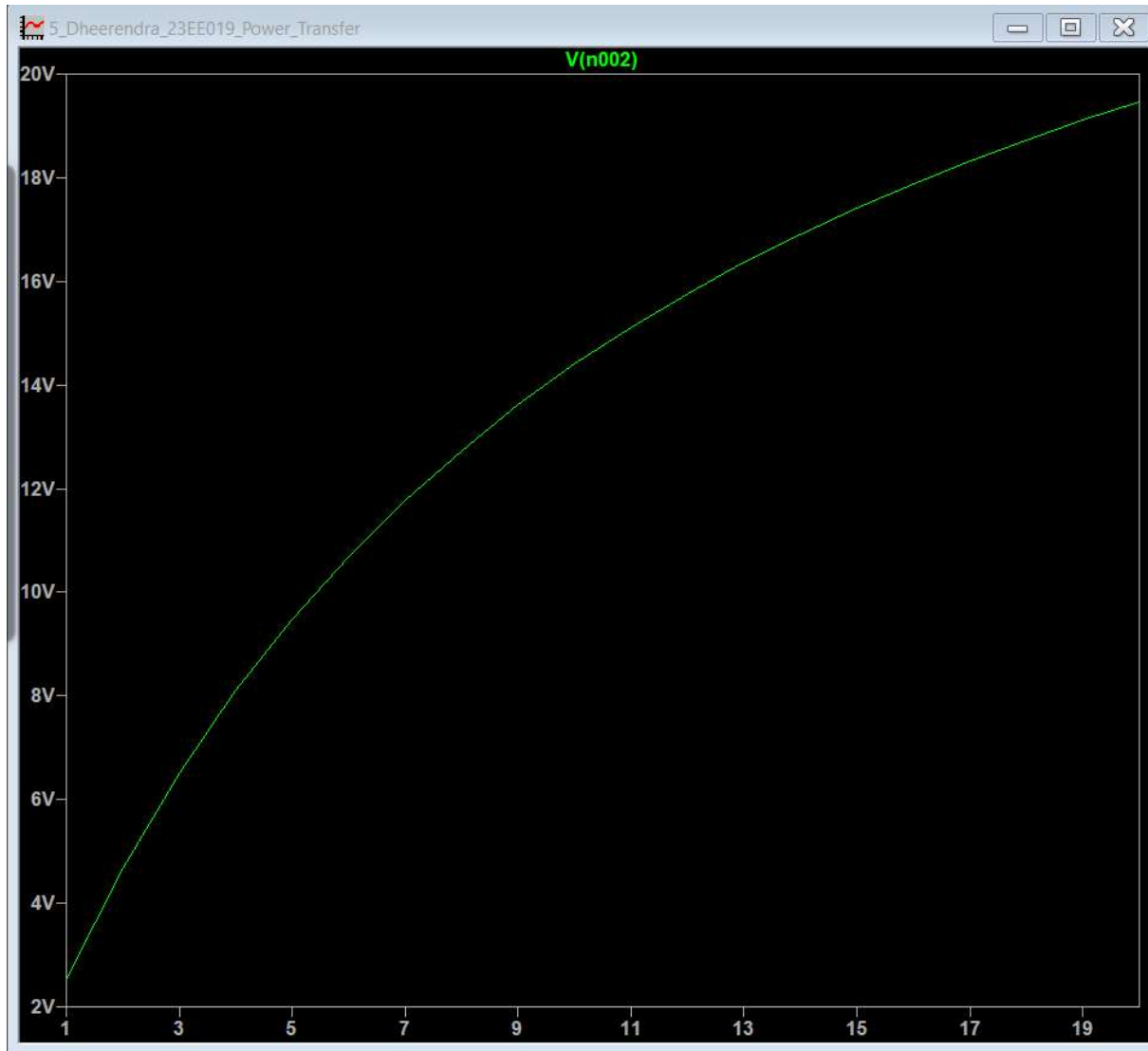
Que5. Determine the maximum Power that can be delivered to the variable resistor R in the circuit



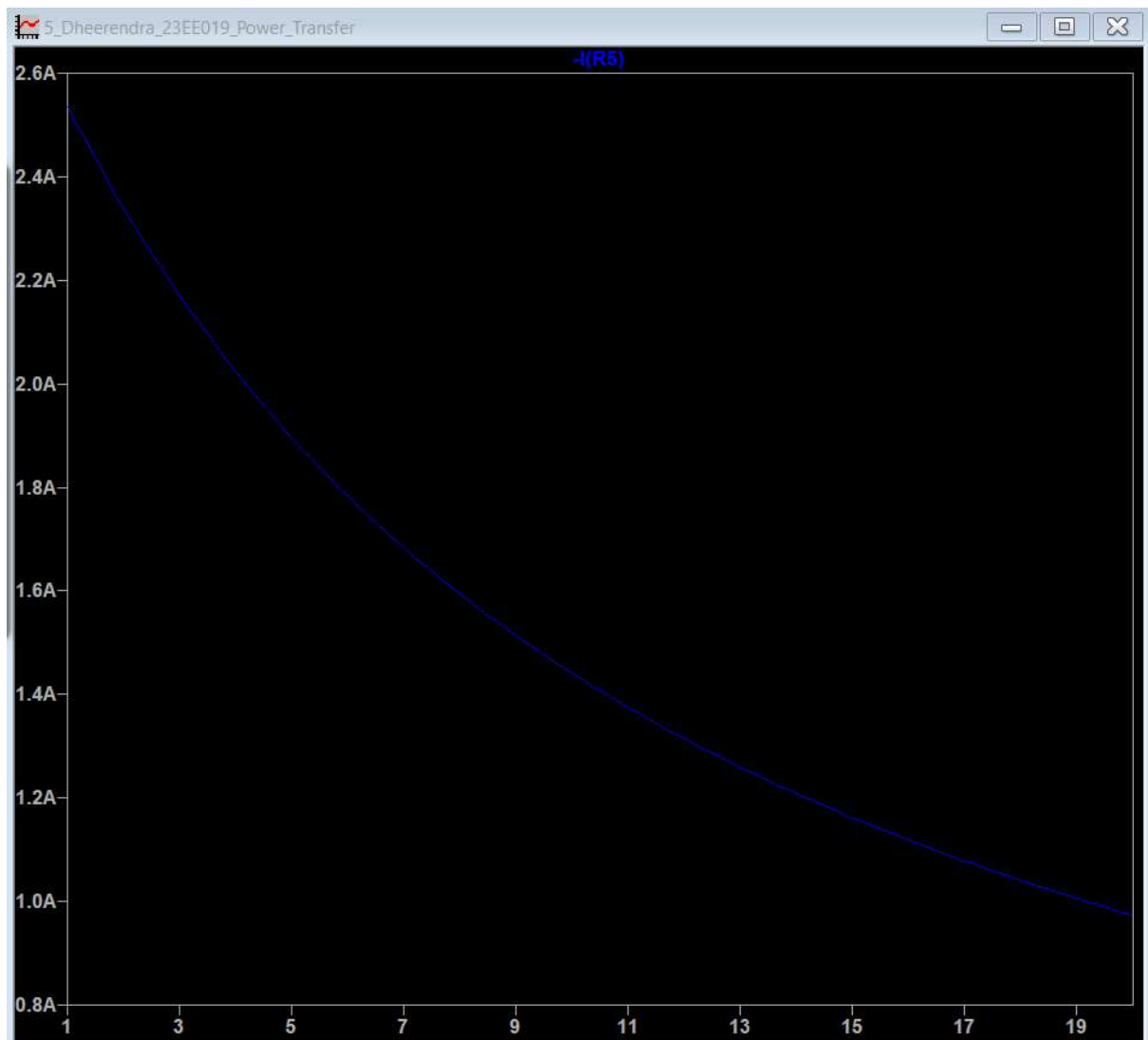
.step param R 1 20 1

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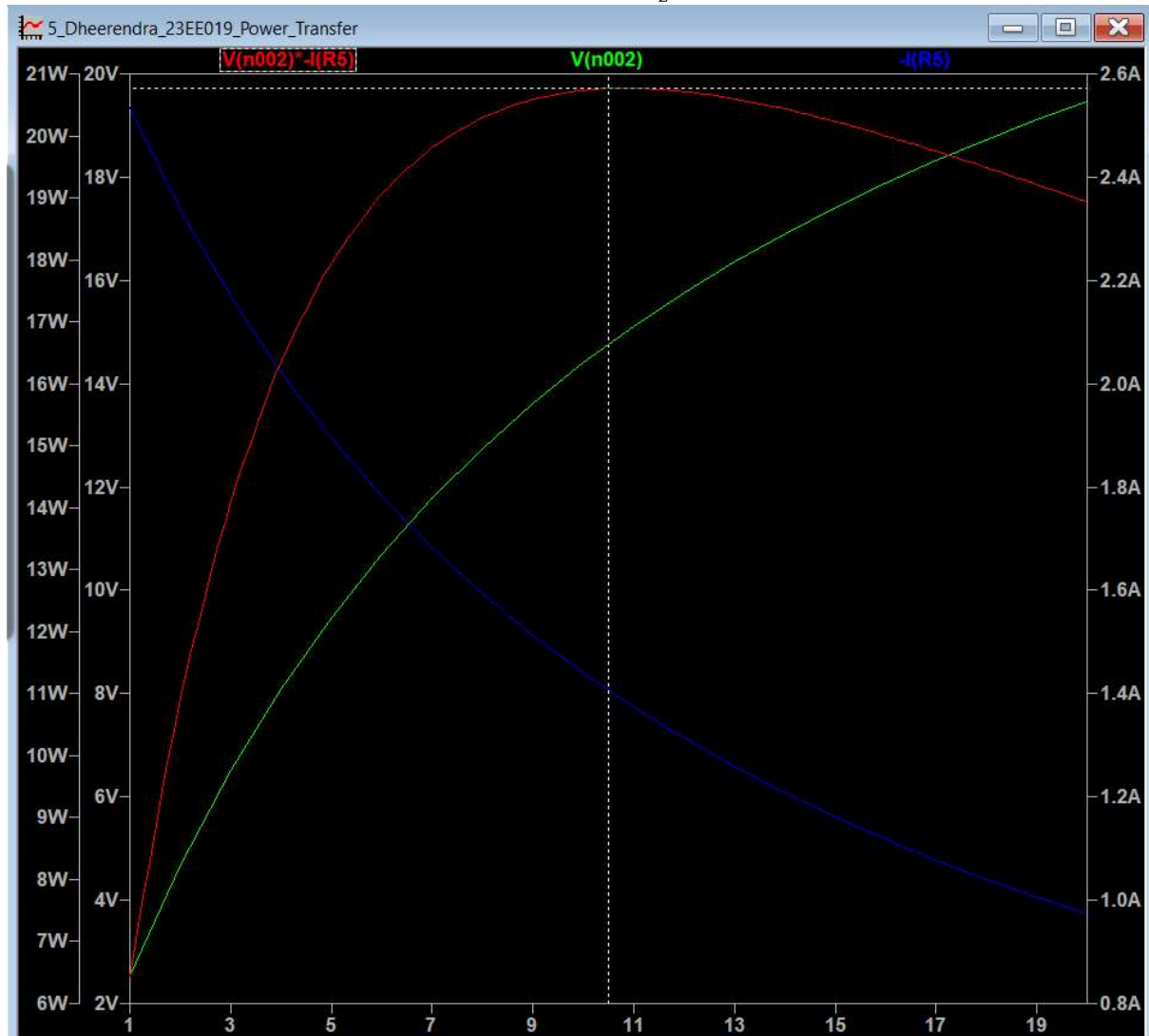
Plot of Voltage across R_L Voltage:



Plot of Current across R_L :



Determination of Maximum Power delivered to load R_L :



Observation Table:

S. No.	V_{th} (V)	$R_L = R_{th}$ (Ω)	I_L (A)	Power through Load Resistance $P_{L(max)} = I_L^2 \cdot R_L$ (W)	
5.	30	10.83	1.385	Practical Value	Theoretical Value
				20.763792	20.775

Results and Conclusion:

As we see that the practical value of maximum Power Transfer through the load Resistance is approximately equal to the theoretical Calculation, hence the maximum Power Transfer Theorem is Verified